

1. $AC = AB + BC = 1.8 + 2.4 = 4.2$ **mi.**
2. $AC = 14$. Also $AB + BC = (2x + 1) + (3x - 5) = 14 \Rightarrow 5x - 4 = 14 \Rightarrow x = \boxed{18/5 = 3.6}$.
Then $AB = 2(3.6) + 1 = 8.2$, so $B = A + AB = -5 + 8.2 = \boxed{3.2}$.
3. $P = A + \frac{3}{4}(B - A) = -6 + \frac{3}{4}(16) = -6 + 12 = \boxed{6}$.
4. $P = A + \frac{3}{3+5}(B - A) = 2 + \frac{3}{8} \cdot 16 = \boxed{8}$.
5. $B - A = (12, -12)$, so $P = A + \frac{3}{5}(B - A) = (-3, 5) + \frac{3}{5}(12, -12) = (-3 + 7.2, 5 - 7.2) = \boxed{(4.2, -2.2)}$.
6. Center is midpoint $((-4 + 6)/2, (2 + 2)/2) = (1, 2)$; $r = \frac{1}{2} \cdot |EF| = \frac{1}{2} \cdot 10 = 5$. $C = 2\pi r = \boxed{10\pi}$, $A = \pi r^2 = \boxed{25\pi}$.
7. $d = \sqrt{(10 + 6)^2 + (-2 - 4)^2} = \sqrt{16^2 + (-6)^2} = \sqrt{256 + 36} = \boxed{\sqrt{292} = 2\sqrt{73}}$. $M = (2, -1)$.
8. $\ell = |9 - 1| = 8$, $w = |2 - (-4)| = 6$. $P = 2(8 + 6) = \boxed{28}$; $A = 8 \cdot 6 = \boxed{48}$.
9. Diagonal $d = \sqrt{(8)^2 + (8)^2} = 8\sqrt{2}$, side $s = d/\sqrt{2} = 8$. $P = \boxed{32}$, $A = \boxed{64}$.
10. $A_{\triangle} = \frac{1}{2} \cdot 8 \cdot 6 = \boxed{24}$. $AC = \sqrt{(5 - 0)^2 + (6 - 0)^2} = \boxed{\sqrt{61}}$.
11. $x + (x + 18) = 90 \Rightarrow 2x = 72 \Rightarrow \boxed{x = 36}$; angles $36^\circ, 54^\circ$.
12. $(3x + 15) + (5x - 3) = 180 \Rightarrow 8x + 12 = 180 \Rightarrow x = \boxed{21}$; angles 78° and 102° .
13. $2y + 14 = 5y - 13 \Rightarrow 3y = 27 \Rightarrow y = \boxed{9}$; angle $= 2(9) + 14 = \boxed{32^\circ}$ (both equal).
14. Bisector $\Rightarrow x + 20 = 2x - 10 \Rightarrow x = \boxed{30}$. Then each half $= 50^\circ$, so $\angle ABC = \boxed{100^\circ}$.
15. Reflect in x -axis: $(x, y) \mapsto (x, -y)$. $A'(3, -1)$, $B'(7, -4)$, $C'(1, -5)$. Perimeter/area preserved (same as original).
16. Translate: $(-2, 3) \rightarrow (3, -1)$; rotate 90° CCW: $(3, -1) \mapsto (1, 3)$. So $P'' = \boxed{(1, 3)}$.
17. Rotate first: $(-2, 3) \mapsto (-3, -2)$; then translate $(+5, -4) \rightarrow (2, -6)$. Not the same as Q16 $\Rightarrow$ order matters.
18. $U(4, -1) \mapsto (-1, -4)$ is $(y, -x)$, which is 270° CCW. So $\boxed{270^\circ \text{ CCW}}$.
19. $r = \sqrt{12^2 + 5^2} = 13$. $C = \boxed{26\pi}$, $A = \boxed{169\pi}$.
20. Sides: $PQ = 6$, $QR = 8$, $PR = 10$ (3-4-5 scaled). Right angle at Q . $P = 6 + 8 + 10 = \boxed{24}$, $A = \frac{1}{2} \cdot 6 \cdot 8 = \boxed{24}$.
21. $RS = \sqrt{(7)^2 + (3)^2} = \sqrt{58}$; $ST = \sqrt{(-4)^2 + (-6)^2} = \sqrt{52} = 2\sqrt{13}$; $TR = \sqrt{(3)^2 + (1)^2} = \sqrt{10}$. $P = \boxed{\sqrt{58} + 2\sqrt{13} + \sqrt{10}}$.

22. $D(0, 6)$. $A = 8 \cdot 6 = \boxed{48}$; $P = 2(8 + 6) = \boxed{28}$.
23. Slopes $PQ = \frac{0-2}{6+1} = -\frac{2}{7}$, $QR = \frac{-2-0}{7-3} = -\frac{2}{4} = -\frac{1}{2}$. Product $\neq -1$, so use distances: $PQ^2 = 53$, $QR^2 = 20$, $PR^2 = 73$ and $PQ^2 + QR^2 = 73 = PR^2$; right at Q .
24. Base area $= 12 \cdot 6 = 72$. Height $h = V/(12 \cdot 6) = 864/72 = \boxed{12}$ cm. $SA = 2(72 + 12 \cdot 12 + 6 \cdot 12) = 2(72 + 144 + 72) = 2 \cdot 288 = \boxed{576 \text{ cm}^2}$.
25. (a) Open top $\Rightarrow$ no top: $A = \ell w + 2\ell h + 2wh = 18 \cdot 12 + 2 \cdot 18 \cdot 10 + 2 \cdot 12 \cdot 10 = \boxed{216 + 360 + 240 = 816 \text{ cm}^2}$. (b) Ribbon one loop mid-height $\Rightarrow$ perimeter of base: $2(\ell + w) = 2(18 + 12) = \boxed{60 \text{ cm}}$.
26. Equal volumes: Cylinder $\pi(4^2)h = 16\pi h = \frac{1}{3}\pi(4^2) \cdot 18 = \frac{1}{3} \cdot 16\pi \cdot 18 = 96\pi \Rightarrow h = \boxed{6}$. Lateral areas: cylinder $2\pi rh = 2\pi \cdot 4 \cdot 6 = 48\pi$; cone πrs , $s = \sqrt{4^2 + 18^2} = \sqrt{340} \approx 18.44$, so $\pi \cdot 4 \cdot 18.44 \approx 73.8\pi$. Cone's lateral is larger.
27. (a) $h = \sqrt{s^2 - r^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = \boxed{8}$. (b) $SA = \pi r(r + s) = \pi \cdot 6(6 + 10) = \boxed{96\pi \text{ cm}^2}$. (c) $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \cdot 36 \cdot 8 = \boxed{96\pi \text{ cm}^3}$.
28. Label wraps once: circumference $\cdot$ height $= A$. $2\pi r \cdot h = 264$, with $h = 8$. So $2\pi r \cdot 8 = 264 \Rightarrow r = \frac{264}{16\pi} = \boxed{\frac{33}{2\pi} \text{ cm}}$, $h = \boxed{8 \text{ cm}}$.
29. $4\pi r^2 = 324\pi \Rightarrow r^2 = 81 \Rightarrow r = \boxed{9}$ cm. $V = \frac{4}{3}\pi r^3 = \boxed{972\pi \text{ cm}^3}$.
30. (a) Total $SA = a^2 + 2as = 14^2 + 2 \cdot 14 \cdot 13 = \boxed{196 + 364 = 560 \text{ cm}^2}$. (b) $h = \sqrt{s^2 - (a/2)^2} = \sqrt{13^2 - 7^2} = \sqrt{169 - 49} = \sqrt{120} = 2\sqrt{30}$. $V = \frac{1}{3}a^2 h = \frac{1}{3} \cdot 196 \cdot 2\sqrt{30} = \boxed{\frac{392}{3}\sqrt{30} \text{ cm}^3}$.
31. $V = 960\pi$. A: $960\pi = \pi \cdot 6^2 h \Rightarrow h = \frac{960}{36} = \boxed{26.\bar{6}}$ cm; lateral $= 2\pi \cdot 6 \cdot h = \boxed{320\pi}$. B: $960\pi = \pi \cdot 8^2 h \Rightarrow h = \frac{960}{64} = \boxed{15}$ cm; lateral $= 2\pi \cdot 8 \cdot 15 = \boxed{240\pi}$. Can B use less lateral area.
32. Cone $V = \frac{1}{3}\pi \cdot 5^2 \cdot 12 = 100\pi$. Cylinder $\pi \cdot 5^2 \cdot H = 25\pi H = 100\pi \Rightarrow H = \boxed{4 \text{ cm}}$.
33. (a) $V = \pi r^2 h + \frac{2}{3}\pi r^3 = \pi \cdot 6^2 \cdot 10 + \frac{2}{3}\pi \cdot 6^3 = 360\pi + 144\pi = \boxed{504\pi \text{ cm}^3}$. (b) Exterior $SA =$ cylinder side $2\pi rh +$ hemisphere $2\pi r^2 +$ bottom $\pi r^2 = 2\pi \cdot 6 \cdot 10 + 2\pi \cdot 36 + \pi \cdot 36 = \boxed{120\pi + 72\pi + 36\pi = 228\pi \text{ cm}^2}$.
34. Nearest 0.01 $\Rightarrow$ interval $\boxed{[7.835, 7.845]}$ m.
35. Sum $= 12.395 \rightarrow$ tenths $\Rightarrow \boxed{12.4}$. Product $= 0.1728$ with 3 and 2 s.f. $\Rightarrow \boxed{0.17}$.
36. Max abs error ≤ 0.5 m; percent $\leq \frac{0.5}{142} \approx \boxed{0.352\%}$.

37. (i) $\rho = 12.0/4.0 = 3.0 \text{ g/cm}^3$ (2 s.f.). (ii) $m \in [11.95, 12.05)$, $V \in [3.95, 4.05)$, so $\rho \in \left[\frac{11.95}{4.05}, \frac{12.05}{3.95}\right) \approx [2.95, 3.05) \text{ g/cm}^3$.